Deep Learning

Security Analytics COSC 4397/6346

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Note: Resources and references are listed on the last slide.

Representation Learning

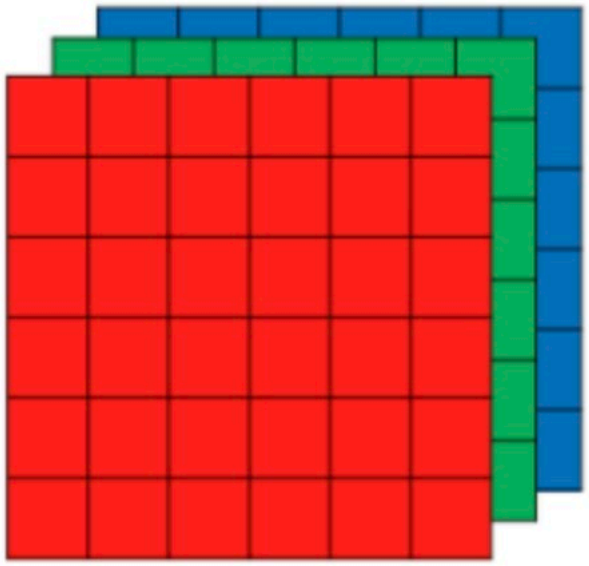
- Idea proposed in “Deep Learning” by LeCun et. al., published in Nature Journal, 2015
- Conventional machine learning algorithms cannot handle data in their raw form
  - Feature engineering and preprocessing steps before training
- Representation learning methods
  - Raw data is fed in to discover better representations
  - Used for classification or recognition tasks

Deep Learning

- Is a form of “Representation Learning” with multiple levels of representation learning
- A Deep Neural Network is composed of:
  - Multiple levels of Representation
  - Each level has simple but non-linear modules
  - Each module contributes to transformation of the raw data to a more abstract representation
- End-to-End learning concept

Convolutional Neural Network (ConvNets)

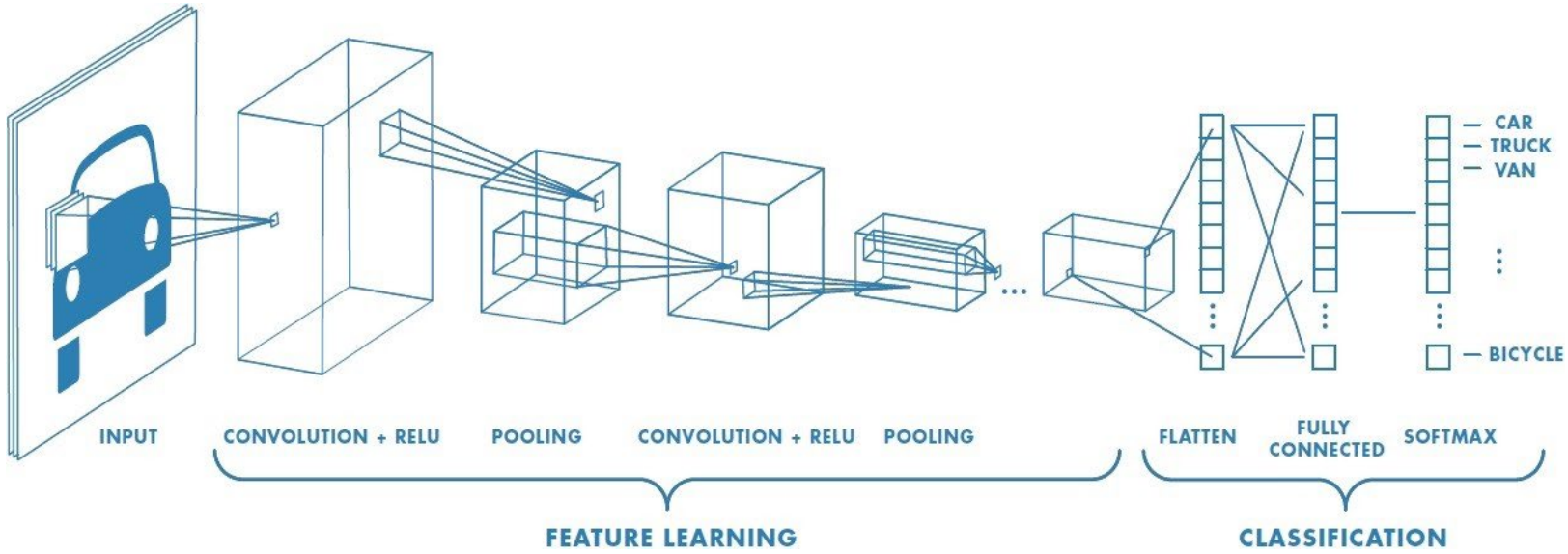
- Supervised Learning Algorithm
- Process data that come in the form of different arrays
- Used most commonly for image classification
  - Sees image as 3D arrays containing pixel intensities
  - Based on image resolution: $h \times w \times d$
  - A color image is of $6 \times 6 \times 3$ array of matrix of RGE



6 x 6 x 3

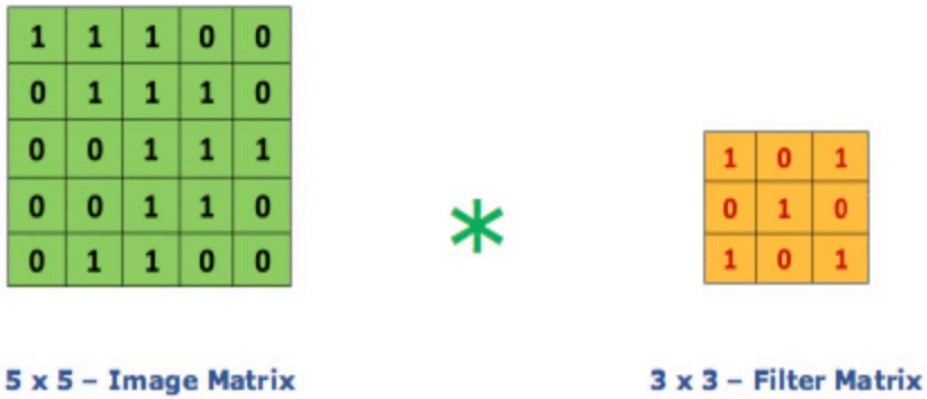
Convolutional Neural Networks (CNNs)

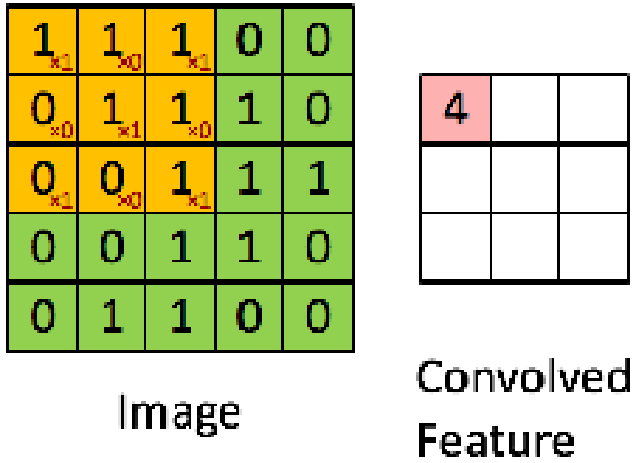
- Stages in a ConvNet
  - Convolution
  - ReLU
  - Pooling
  - Fully Connected
  - SoftMax



Convolution Layer and ReLU

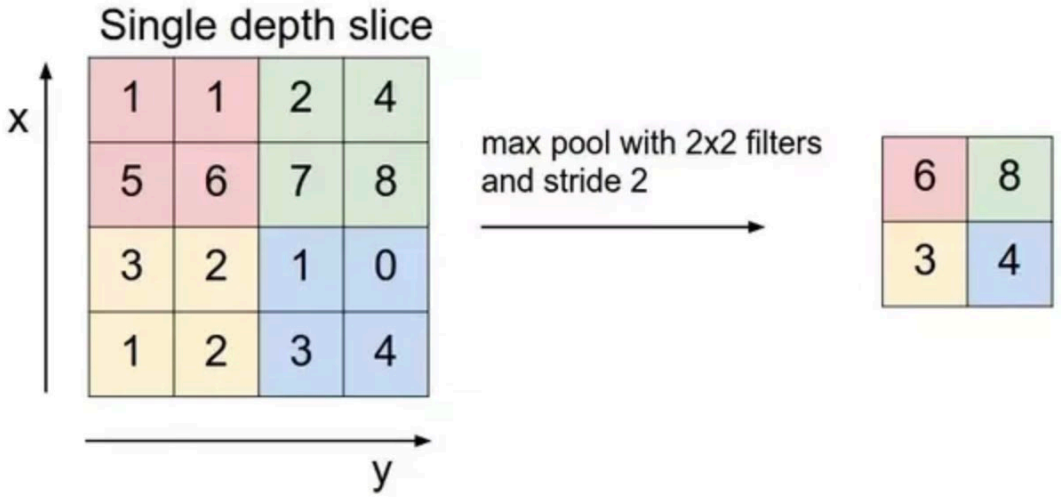
- Main building block of a CNN
- Creation of a feature map
  - Input * Convolution filter → Feature Map
  - $h \times w \times d * f_h \times f_w \times f_d$
  - Feature Map: $(h - f_h + 1) \times (w - f_w + 1) \times 1$
- Rectified Linear Unit (ReLU)
  - Introduce non-linearity in our CNN
    - $f(x) = \max(0, x)$
  - Other similar functions, tanh or sigmoid





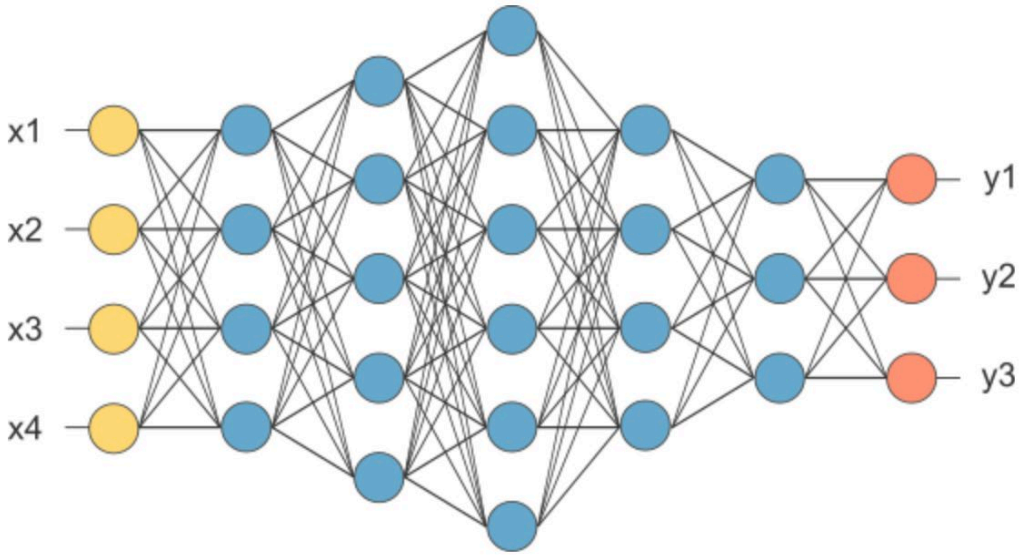
Pooling Layer

- Reduces the number of parameters in case of large inputs
- Spatial pooling reduces dimensionality of each map without loss of information
- Types
  - Max Pooling
  - Average Pooling
  - Sum Pooling
- Advantages:
  - shortens the training time
  - combats overfitting



Fully Connected Layer and SoftMax

- *Flatten*: Convert the outputs of the convolution and pooling layers from 3D to 1D
- Fully Connected Layer acts as a classifier
- SoftMax: Final Activation Layer to output the probabilities

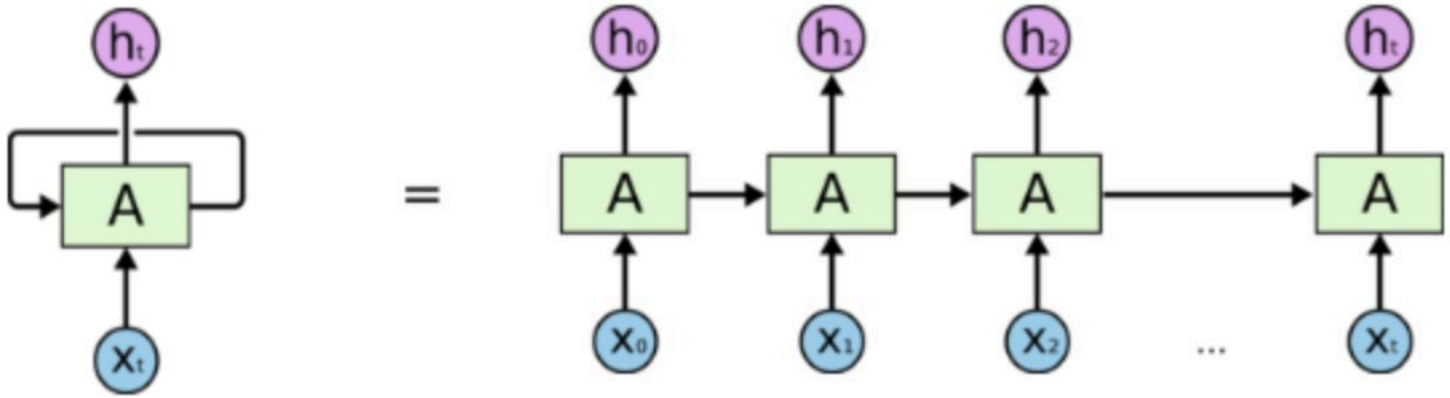


Shortcomings of a CNN

- Accept a fixed size vector as input and output
- How do we learn a representation of a sequential input?

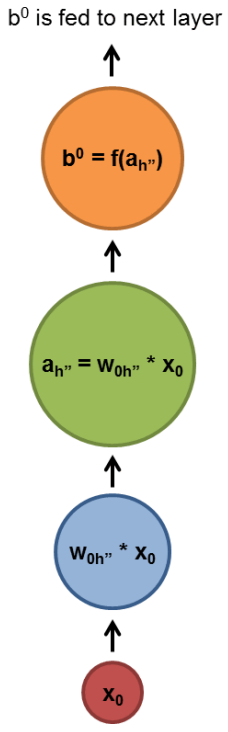
Recurrent Neural Networks (RNNs)

- Human thoughts have persistence
- RNNs take into account context in their inputs
- Multiple copies of the same network
- The message passes from one unit to its successor



Recurrent Neural Networks (RNN)

- Input vector x and output vector y
- Current state, $h_t = f(h_{t-1}, x_t)$
- With activation function, tanh
  - $h_t = tanh(W_{hh}h_{t-1} + W_{xh}x_t)$
- Final output, $y = (W_{hy}h_t)$
- Three weight matrices
  - W_{hh}: Previous Hidden State
  - W_{xh}: Current Input
  - W_{hy}: Between Hidden State and Output

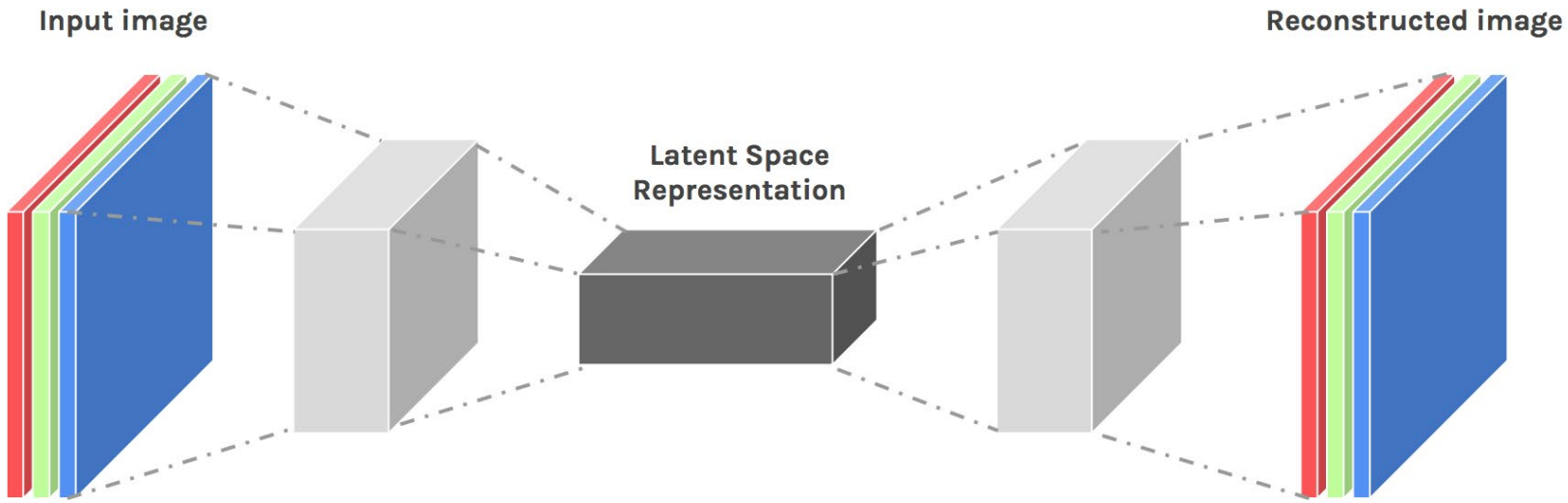


Applications and Types of RNN

- Applications
  - Language Modelling and Text Generation
  - Machine Translation
  - Speech Recognition
  - Generation of Image Captions
  - Video Tagging
- Types
  - LSTM
  - GRU

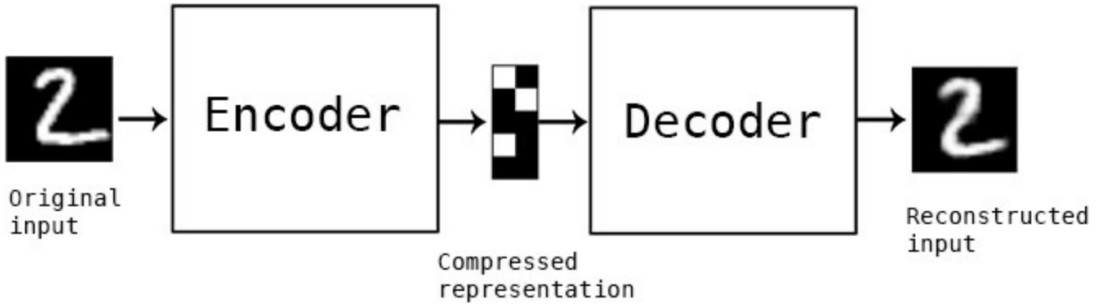
Simple Auto-encoders

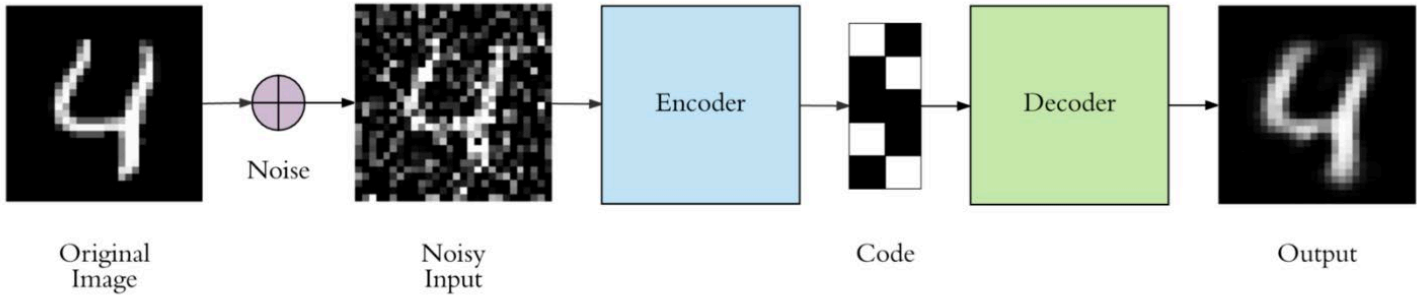
- Generative model used for Unsupervised applications
- Input is same as the output



Simple Auto-encoders

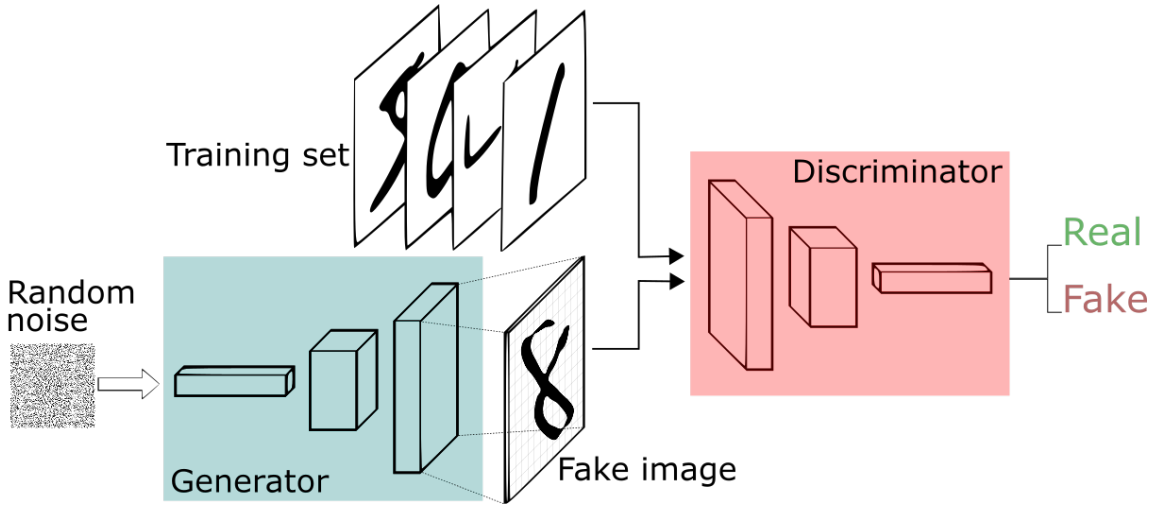
- Learn features from an input and reconstruct the output from given image
- Features are called *latent representation* of the input
- Properties of an Autoencoder
  - Data-specific
  - Lossy
  - Unsupervised/self-supervised
- Parts of an Autoencoder Architecture
  - Encoder
  - Decoder
- Not as effective as GANs





Generative Adversarial Networks

- Two players performing two entirely different roles
  - Generator
    - Produce new data from noise
    - Maximize the probability of fooling discriminator
  - Discriminator
    - Acts as the judge
    - Decides between real and generated
    - Guides the generator



- In equilibrium (depending on goal)
  - Generator perfectly captures the input data distribution
  - Discriminator is confused

Resources and References

- Deep Learning, Nature Journal, 2015
- Getting Started with Keras, https://keras.io/
- Working with Google Colab, https://colab.research.google.com/notebooks/welcome.ipynb
- Some blogs:
  - https://medium.com/@RaghavPrabhu/understanding-of-convolutional-neural-network-cnn-deep-learning-99760835f148
  - https://hackernoon.com/rnn-or-recurrent-neural-network-for-noobs-a9afbb00e860
  - https://towardsdatascience.com/autoencoders-made-simple-6f59e2ab37ef
  - https://medium.freecodecamp.org/an-intuitive-introduction-to-generative-adversarial-networks-gans-7a2264a81394